MATH 2211, SPRING 2026: HOMEWORK 2

(1) Can you use the argument from problem 2 on Homework 1 to show that there are infinitely many primes that are congruent to 1 mod 4? What goes wrong?

(2) Use strong induction to prove the following fact: Every integer $n \geq 12$ can be written in the form $n = 4a + 5b$ for $a, b \in \mathbb{N}$.
Hint: You might need multiple base cases.

(3) Suppose that F is a field. Verify the following facts using only the defining conditions for a field given in Lecture 4:
  (a) For any $a \in F$, $0 \cdot a = 0$.
  (b) For any $a \in F$, $(-1) \cdot a = -a$.
  (c) If $a, b \in F$ are such that $a \cdot b = 0$, then either $a = 0$ or $b = 0$.

(4) Solve the following linear system of equations in the field $\mathbb{Z}/5\mathbb{Z}$:

$$2x + 3y + z = 1$$
$$4x + 2y + 3z = 2$$

(5) Compute the real and imaginary parts of $\frac{\pi+i}{5-i} \in \mathbb{C}$.

(6) Fix an integer $n \geq 1$ and set $\zeta = e^{2\pi i/n} \in \mathbb{C}$.
  (a) Show that the product $\zeta \cdot \zeta \cdots \zeta^{n-1}$ is equal to ± 1. When is it 1 and when is it -1?
  (b) Show that $1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1} = 0$.